

Final Project Report Template

1. Introduction

1.1. Project overviews

The **Food Ordering Behaviour and Consumer Trends** project is designed to analyze customer food ordering patterns and restaurant performance using Tableau. The project focuses on transforming raw restaurant order data into meaningful business insights through data preparation, visualization, interactive dashboards, and Tableau Stories.

The analysis includes customer preferences across different cuisines, meal types, restaurant types, customer ratings, city-wise order distribution, repeat ordering behaviour, and average order value. Interactive filters, parameters, and Key Performance Indicators (KPIs) are incorporated to enable users to explore the data dynamically and make informed decisions.

The project demonstrates the complete business intelligence workflow, starting from data collection and preprocessing to visualization development, dashboard design, story creation, and project documentation. The insights generated can help restaurant owners and food delivery businesses understand customer behaviour, identify market trends, improve customer satisfaction, and support data-driven business decision-making.

1.2. Objectives

- To analyze customer food ordering behaviour using Tableau.
- To identify the most popular cuisines and meal types.
- To compare restaurant performance based on order volume and average order value.
- To analyze customer ratings across different meal types.
- To identify the top-performing cities based on food orders.
- To study repeat ordering behaviour.
- To develop an interactive dashboard and Tableau Story for business insights.

2. Project Initialization and Planning Phase

2.1. Define Problem Statement

The project focuses on analyzing food ordering behaviour and consumer trends from the perspective of different stakeholders. Restaurant owners and food delivery businesses generate large amounts of order-related data, but understanding customer preferences, popular cuisines, meal types, ratings, revenue trends, and city-wise ordering patterns from this data can be difficult when analyzed manually.

The identified customer problem statements are:

Problem Statement (PS)	I am (Customer)	I'm trying to	But	Because	Which makes me feel
PS-1	A Restaurant Owner	understand customer ordering patterns and identify the most popular cuisines and meal types	I cannot easily analyze customer order data	because the data is large and difficult to interpret manually	uncertain while making menu planning and business decisions
PS-2	A Food Delivery Manager	improve customer satisfaction and delivery performance	I cannot quickly identify customer preferences and delivery trends	because the available data is not presented in a clear and visual format	frustrated as it affects service quality and decision-making

Therefore, the main problem is the **difficulty in converting large food-ordering datasets into meaningful and easily understandable business insights**. A structured data analysis and visualization approach is required to help stakeholders understand consumer behaviour and make better business decisions.

2.2. Project Proposal (Proposed Solution)

Project Overview

The proposed project, **Food Ordering Behaviour and Consumer Trends: A Structured Analysis of Choices and Habits**, aims to analyze food ordering behaviour and consumer trends using **Tableau**.

Objective

The objective of the project is to analyze food ordering behaviour using Tableau by identifying popular cuisines, customer preferences, meal types, ratings, and city-wise ordering patterns. The project aims to provide interactive visual insights that support better business decision-making.

Scope

The project focuses on analyzing restaurant and food-ordering data using Tableau. It includes data preparation, visualization, dashboard development, and story creation to identify customer behaviour and restaurant performance trends. The project is limited to the provided food-ordering dataset.

Problem Statement

Restaurant owners and food delivery businesses generate large amounts of order data. Analyzing this data manually is difficult, making it challenging to understand customer preferences, popular cuisines, revenue trends, and overall business performance.

Impact

The proposed solution helps businesses make data-driven decisions by providing interactive dashboards and visual insights. It supports menu planning, marketing strategies, customer satisfaction, and operational improvements.

Proposed Solution

The proposed solution is to use **Tableau** to connect, clean, analyze, and visualize the food-ordering dataset. Interactive charts, filters, KPIs, parameters, dashboards, and stories are developed to understand customer ordering behaviour and restaurant performance effectively.

Approach

The project uses Tableau to connect, clean, and analyze the restaurant dataset. Various charts, filters, KPIs, parameters, dashboards, and stories are created to visualize customer ordering behavior and restaurant performance effectively.

Key Features

- Interactive Tableau Dashboard
- Customer Order Trend Analysis
- Cuisine Preference Analysis
- City-wise Order Analysis
- Meal Type Analysis
- Customer Rating Analysis

- Revenue Insights
- Interactive Filters and Parameters
- Tableau Story for Business Insights

Resource Requirements

Hardware Requirements:

- Intel Core i3 / Ryzen 3 or higher
- Dual-Core Processor
- 8 GB RAM
- Minimum 10 GB free storage space

Software Requirements:

- Tableau Desktop as the development environment

Data Requirements:

- Food Ordering Dataset
- CSV format

2.3. Initial Project Planning

The project was planned using a sprint-based approach. The work was divided into functional requirements and user stories so that data collection, loading, preparation, visualization, dashboard/story development, documentation, and demonstration could be completed systematically.

Sprint	Functional Requirement (Epic)	User Story Number	User Story / Task	Story Points	Priority	Team Members	Sprint Start Date	Sprint End Date (Planned)
Sprint-1	Dataset Collection	USN-1	As a data analyst, I want to download the restaurant dataset so that I can start the project.	2	High	Nikita Nitin Patil	10-08-2026	10-08-2026
Sprint-1	Data Loading	USN-2	As a data analyst, I want to load the dataset into Tableau for analysis.	1	High	Nikita Nitin Patil	11-08-2026	11-08-2026
Sprint-1	Data Preparation	USN-3	As a data analyst, I want to clean and prepare the dataset by handling data types and missing values to ensure accurate analysis.	3	Medium	Nikita Nitin Patil	12-08-2026	13-08-2026
Sprint-1	Data Visualization	USN-4	As a data analyst, I want to create visualizations to identify customer ordering trends and business insights.	5	High	Nikita Nitin Patil	15-08-2026	19-08-2026
Sprint-2	Dashboard & Story Development	USN-5	As a business user, I want an interactive dashboard and story so that I can easily understand business insights.	5	High	Nikita Nitin Patil	20-08-2026	22-08-2026
Sprint-2	Publishing, Documentation & Demo Recording	USN-6	As a data analyst, I want to publish the Tableau project, complete the documentation, and record the demo video for successful project submission.	8	High	Nikita Nitin Patil	23-08-2026	26-08-2026

The sprint plan provides a structured sequence for completing the project from dataset collection through visualization, dashboard and story development, documentation, and final demonstration.

3. Data Collection and Preprocessing Phase

3.1. Data Collection Plan and Raw Data Sources Identified

Data Collection Plan

The dataset was downloaded from the **SkillWallet Project Workspace** after logging into the SkillWallet portal. The downloaded dataset was then imported into **Tableau Desktop** for data preparation, visualization, dashboard development, and story creation.

Raw Data Sources Identified

The project uses a restaurant and food-ordering dataset downloaded from the SkillWallet Project Workspace. The dataset contains information related to:

- Customer orders
- Cuisines
- Meal types
- Customer ratings
- Restaurant types
- City details

- Revenue
- Delivery information

This information is used to analyze customer behaviour, ordering patterns, restaurant performance, and other business-related insights.

Source Name	Description	Location /URL	Format	Size	Access Permissions
food_ordering_behavior_dataset	Contains customer orders, cuisines, meal types, customer ratings, restaurant types, city details, revenue, and delivery information used for business analysis.	SkillWallet Project Workspace (Downloaded Dataset)	CSV	5.07 MB	SkillWallet Project Access

3.2. Data Quality Report

The Data Quality Report was prepared to identify and assess potential data quality issues in the selected dataset before performing analysis and visualization. The dataset was reviewed for missing values, duplicate records, incorrect data types, and inconsistent categorical values.

Data Source	Data Quality Issue	Severity	Resolution Plan
food_ordering_behavior_dataset	The dataset was checked for missing values, duplicate records, incorrect data types, and inconsistent categorical values before analysis. No major data quality issues affecting the analysis were identified.	Low	The dataset was validated by assigning appropriate data types in Tableau, verifying data consistency, and using the cleaned dataset for visualization, dashboard development, and story creation.

Data Quality Assessment

The dataset was reviewed before analysis to ensure that the available data was suitable for visualization and business analysis. The assessment focused on:

- Missing values
- Duplicate records
- Incorrect data types
- Inconsistent categorical values

No major data quality issues affecting the analysis were identified. Appropriate data types were assigned in Tableau, and data consistency was verified before using the dataset for visualization, dashboard development, and story creation.

3.3. Data Exploration and Preprocessing

The data exploration and preprocessing phase was performed to understand the dataset, identify potential data-quality issues, and prepare the data for reliable analysis and visualization in Tableau.

Data Overview

The project uses the **food_ordering_behavior_dataset** downloaded from the SkillWallet Project Workspace. The dataset is available in **CSV format** with a size of **5.07 MB**. It contains information such as customer orders, cuisines, meal types, customer ratings, restaurant types, cities, revenue, delivery time, and other attributes used for restaurant

analytics.

Data Cleaning

The dataset was reviewed to identify missing values, duplicate records, incorrect data types, and inconsistent categorical values. Data consistency was verified, and the dataset was prepared for analysis in Tableau to ensure reliable visualizations and business insights.

Data Transformation

The dataset was transformed in Tableau by applying filters, sorting data, grouping relevant fields, and creating calculated fields where required for visualization and dashboard analysis.

Data Type Conversion

Appropriate data types such as **String, Number, and Date** were verified and assigned to the dataset fields in Tableau to ensure accurate analysis and calculations.

Column Splitting and Merging

No column splitting or merging was required because the dataset structure was already suitable for analysis in Tableau.

Data Modeling

The project used a **single dataset**; therefore, no table relationships or joins were required. Measures and dimensions available in the dataset were used to create visualizations, dashboards, and stories in Tableau.

Save Processed Data

The cleaned and prepared dataset was saved within the **Tableau workbook** and used to create interactive visualizations, dashboards, and stories for business analysis.

4. Data Visualization

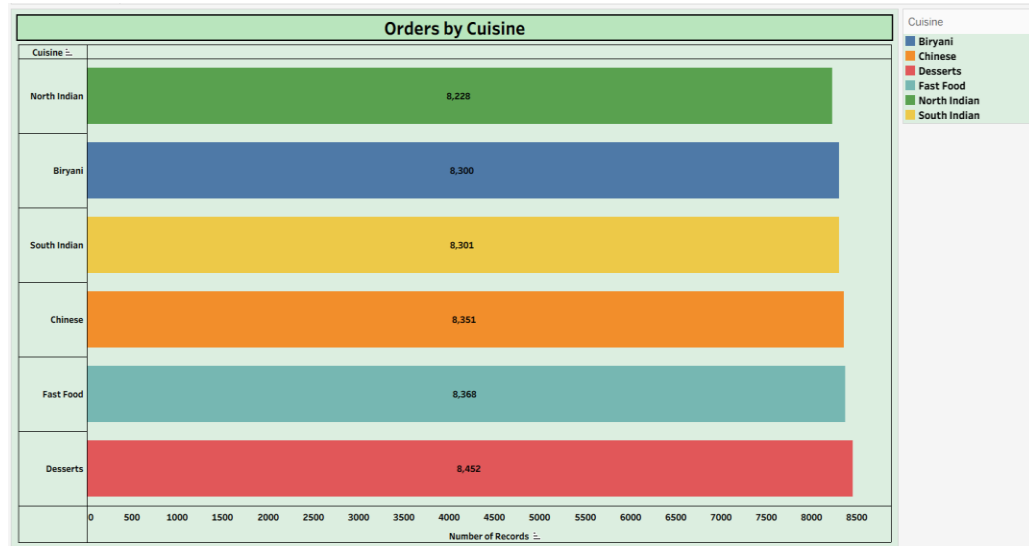
4.1. Framing Business Questions

- 1) Which cuisines receive the highest number of customer orders?
- 2) How do food orders vary across different age groups and restaurant types?
- 3) Which restaurant types receive the highest number of customer orders?
- 4) How do company types influence meal type orders?
- 5) How are customer ratings distributed across different meal types?
- 6) Which cities have the highest food order volume?
- 7) What is the distribution of repeat orders for the selected cuisine?
- 8) Which restaurant types have the highest average order value?

4.2. Developing Visualizations

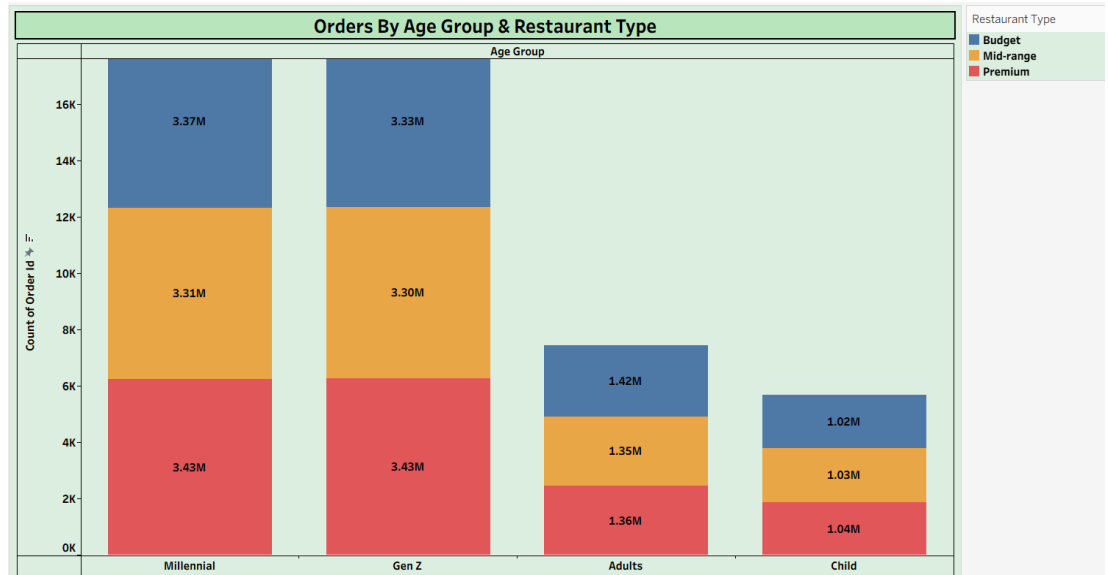
Business Question 1

- **Visualization:** Horizontal bar chart showing the total number of customer orders for each cuisine.
- **Screenshot of Visualization:**



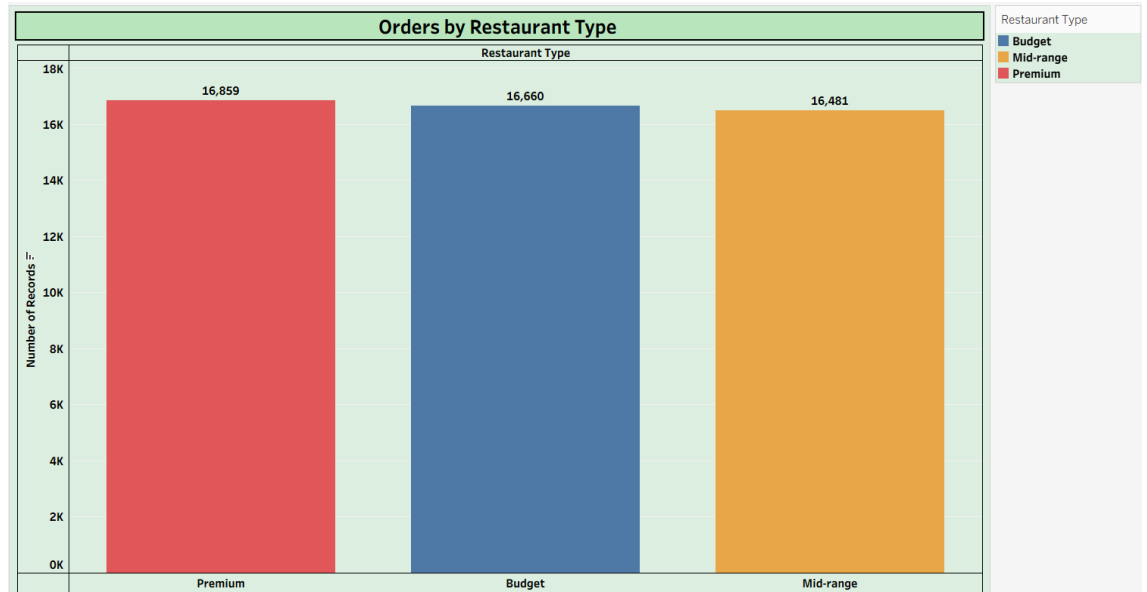
Business Question 2

- **Visualization:** Stacked bar chart showing the distribution of food orders across different age groups and restaurant types.
- **Screenshot of Visualization:**



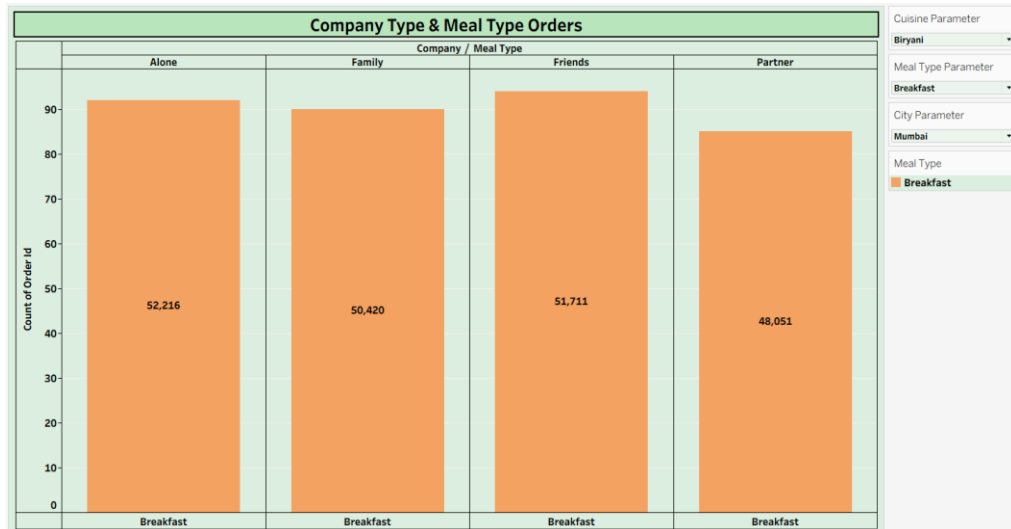
Business Question 3

- **Visualization:** Bar chart comparing the total number of customer orders across different restaurant types.
- **Screenshot of Visualization:**



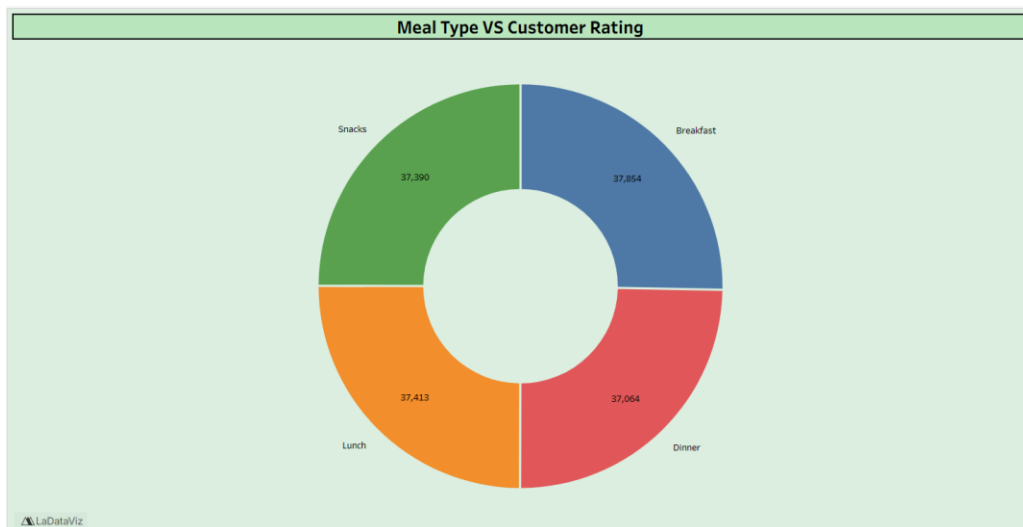
Business Question 4

- **Visualization:** Bar chart showing the distribution of meal type orders across different company types.
- **Screenshot of Visualization:**



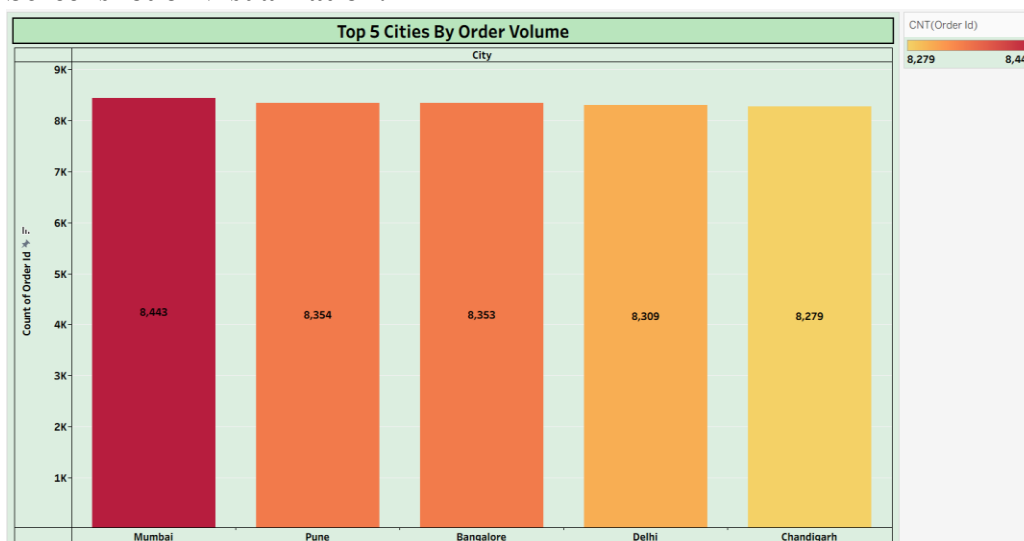
Business Question 5

- **Visualization:** Donut chart showing the distribution of customer ratings across different meal types.
- **Screenshot of Visualization:**



Business Question 6

- **Visualization:** Bar chart showing the top five cities based on the total number of food orders.
- **Screenshot of Visualization:**



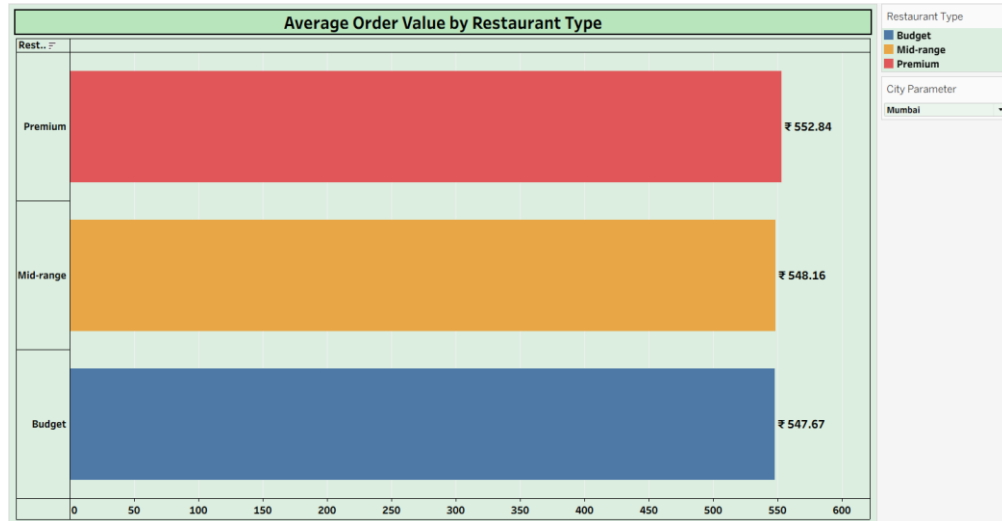
Business Question 7

- **Visualization:** Bar chart showing the distribution of repeat and non-repeat orders for the selected cuisine.
- **Screenshot of Visualization:**



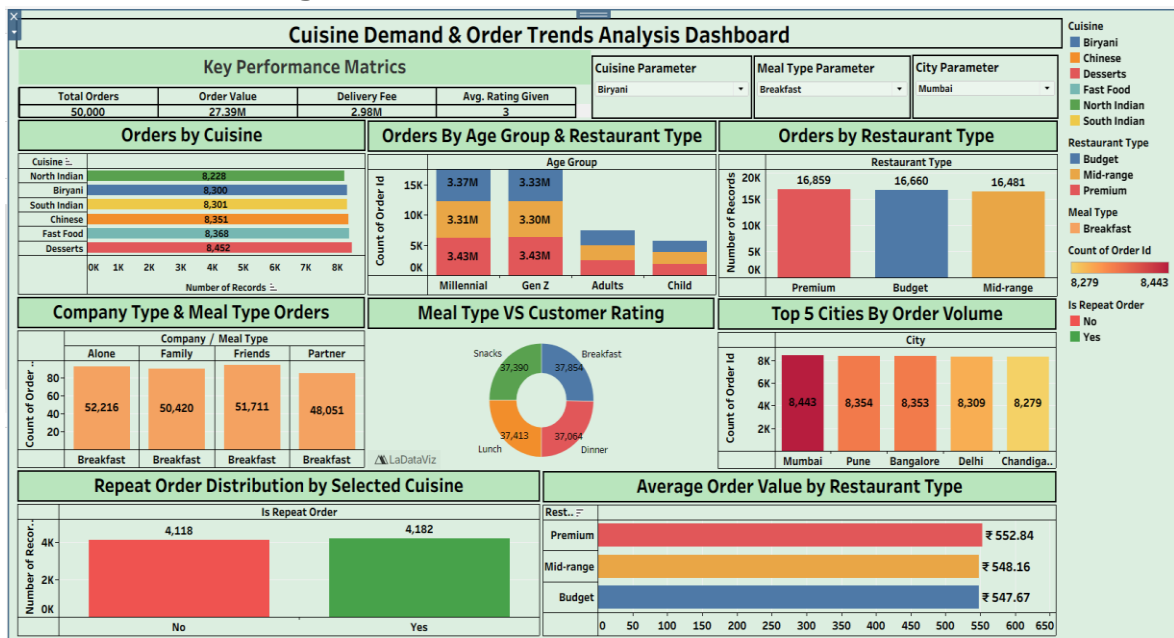
Business Question 8

- **Visualization:** Horizontal bar chart comparing the average order value across different restaurant types.
- **Screenshot of Visualization:**



5. Dashboard

5.1. Dashboard Design File



Major Outcomes

- **The dashboard analyzes 50,000 customer orders**, providing a comprehensive overview of food ordering behaviour and consumer trends.
- **Desserts received the highest number of customer orders**, indicating a strong customer preference among all cuisines.
- **Premium restaurants recorded the highest order volume**, showing greater customer demand compared to other restaurant types.
- **Mumbai ranked as the top city by order volume**, followed by the other leading cities displayed in the dashboard.
- **Premium restaurants achieved the highest average order value**, indicating higher customer spending.
- **Customer ratings are distributed across different meal types**, helping evaluate customer satisfaction.
- **The repeat order analysis shows the distribution of repeat and non-repeat orders** for the selected cuisine, providing insights into customer loyalty.
- **Interactive filters and parameters for Cuisine, Meal Type, and City** enable users to explore the data dynamically and support better business decision-making.

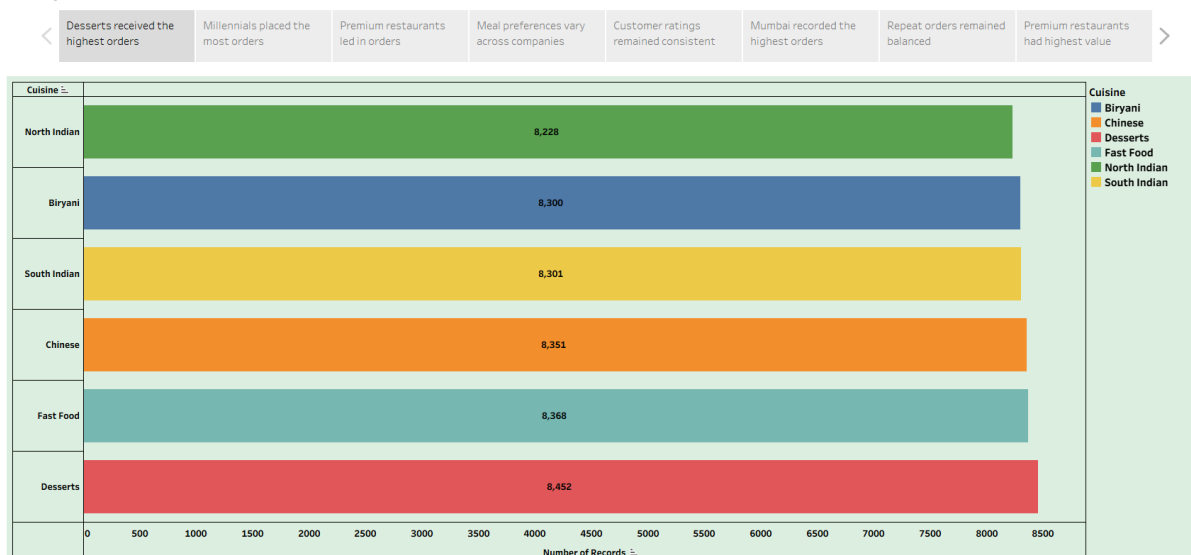
Dashboard Public Link:

https://public.tableau.com/views/Food_Ordering_Dashboard_17874761300210/Dashboard?:language=enUS&publish=yes&:sid=&:redirect=auth&:display_count=n&:origin=viz_share_link

6. Report

6.1. Story Design File

Story



Observations

- **Desserts emerged as the most ordered cuisine**, indicating strong customer preference compared to other cuisines.
- **Premium restaurants recorded the highest number of customer orders**, reflecting greater demand for premium dining options.
- **Mumbai ranked as the leading city in food order volume**, highlighting it as the strongest market among the top five cities.
- **Premium restaurants achieved the highest average order value**, indicating higher customer spending per order.
- **Customer ratings varied across different meal types**, providing insights into customer satisfaction and dining preferences.
- **Repeat order analysis revealed customer loyalty patterns** for the selected cuisine, helping identify repeat purchasing behaviour.
- **Company type influenced meal ordering behaviour**, showing differences in meal preferences across various company categories.
- **The Tableau Story presents a step-by-step analysis of customer ordering behaviour**, enabling users to understand trends and make informed business decisions.

Story Public Link:

https://public.tableau.com/views/Food_Ordering_Story_17874769308540/Story?:language=en-US&publish=yes&:sid=&:redirect=auth&:display_count=n&:origin=viz_share_link

7. Performance Testing

7.1 Utilization of Data filters

Sr. No.	Filter / Parameter Used	Purpose
1	Cuisine Parameter	Analyze data for the selected cuisine.
2	City Parameter	View customer orders by city.
3	Meal Type Parameter	Analyze orders based on meal type.

7.2 No of Calculation Field

Sr. No.	Calculated Field	Purpose
1	Age Group	Categorizes customers into different age groups for analysis.

7.3 No of Visualization

Sr. No.	Visualization Name
1	Orders by Cuisine
2	Orders by Age Group & Restaurant Type
3	Orders by Restaurant Type
4	Company Type & Meal Type Orders
5	Meal Type vs Customer Rating
6	Top 5 Cities by Order Volume
7	Repeat Order Distribution by Selected Cuisine
8	Average Order Value by Restaurant Type

8. Conclusion/Observation

The **Food Ordering Behaviour and Consumer Trends** project successfully analyzed customer

ordering patterns using Tableau. The project transformed raw data into meaningful insights through interactive visualizations, dashboards, and stories. It identified popular cuisines, restaurant performance, city-wise order distribution, customer ratings, repeat ordering behaviour, and average order values. The developed dashboard enables users to explore the data interactively and supports data-driven decision-making for restaurants and food delivery businesses.

9. Future Scope

- Integrate real-time food ordering data for continuous analysis.
- Apply predictive analytics to forecast customer demand and ordering trends.
- Develop personalized food recommendation systems based on customer preferences.
- Expand the dashboard with additional KPIs and advanced business analytics.
- Integrate the dashboard with web or mobile applications for broader accessibility.

10. Appendix

10.1. Source Code(Calculated Field)

Calculated Field Name: Age Group

```
IF [Age] <= 20 THEN "Child"  
ELSEIF [Age] <= 30 THEN "Gen Z"  
ELSEIF [Age] <= 40 THEN "Millennial"  
ELSE "Adults"  
END
```

10.2. GitHub & Project Demo Link

GitHub Repository:

<https://github.com/PatilNikita08/Food-Ordering-Behaviour-and-Consumer-Trends>

Project Demo Video:

https://drive.google.com/file/d/1xPBvsFyN7RTcl6zB2_Ose6Uajh9iDOHg/view?usp=sharing